



ADITHYA SIPKADUWA

ELETRONIC ENGINEER

I am a motivated and enthusiastic individual eager to embark on a fulfilling career path. With a strong willingness to learn and grow, I am committed to acquiring new skills and contributing positively to any team or project. I am eager to take on new challenges and make a meaningful impact in any role I undertake.

ABOUT ME

SIPKADUWE MADUWE GURUGE
ADITHYA SIPKADUWA

+94 750925008

www.linkedin.com/in/adithya-sipkaduwa-a894253b9

adithya.sipkaduwa@outlook.com

464/1, palanwaththa, pannipitiya

TECHNICAL SKILLS

- Embedded Electronic Design & Development
- Electronic Product Design
- Microcontroller Interfacing and Programming : Arduino, ESP32 & NodeMCU, STM32F4, STM32F1, FPGA, Raspberry Pi, Feather, 8051
- IDEs : Arduino IDE, STM32Cube IDE, Xilinx, VS Code
- PCB Design and Schematic Design : KiCad
- Simulation Design and Schematic Design : Proteas
- Antenna Design
- Programming Languages : Assembly, VHDL, C, C++, Java, HTML, PHP, SQL, Python
- UI/UX Design
- Prompt Writing
- Data Analysis
- Computer Networking
- Machine Learning

WORK EXPERIENCES

Trainee - Electronics and Communication Engineer

Communication Engineering Division

Arthur C Clarke Institute for Modern Technologies (ACCIMT)

(Apr 2025 - Aug 2025)

- Developed a Railway Elephant Detection System using Machine Learning and thermal imaging
- Designed and programmed an Environmental Data Logger and a powered Portable Weather Station
- Designed and analyzed PCBs for specialized applications
- Conducted RF cable and antenna testing using Vector Network Analyzers (VNA) and Field Fox RF Analyzers
- Calibrated satellite ground station antennas
- Performed power quality analysis and executed comprehensive capacity and life-cycle testing on industrial and automotive batteries
- Diagnosed and repaired laptop hardware and analyzed pressure sensor sensitivity for consumer electronics using external piping methods

Visiting Technician

Electrical & Computer Engineering Department

The Open University of Sri Lanka - Nawala

(Oct 2024 - Dec 2024)

- Set up and maintained laboratory equipment
- Conducted tests and experiments
- Prepared the laboratory environment for academic activities
- Calibrated, tested, and repaired electronic equipment
- Supervised and supported students during practical sessions

Trainee - IT Assistant (Under Communication Engineering)

Spectrum Management Division

Telecommunications Regulatory Commission of Sri Lanka

(May 2024 - Sep 2024)

- Assisted in frequency allocation and monitored VHF/UHF services using Monitor Plus to ensure compliance with ITU standards
- Participated in training workshops conducted by Dialog 10K Data Centre and Dialog Engineering Head Office
- Supported the Type Approval process for communication devices, verifying technical safety and SAR limits
- Conducted broadcast simulations and interference analysis using CHIRplus_BC software
- Managed IMEI registrations and device verification via the TRCSL IMEI System for legal imports and solved TAC number issues and IMEI number issues using ITU database
- Performed structured cabling tasks, including Ethernet wiring, patch panel terminations, and studied fiber optic transceiver modules (Finisar)

Quality Assurance Technician

Global System Solutions International (Pvt) Ltd

(Sep 2018 - Jan 2019)

SKILLS

- Research & Development (R&D) and Innovation
- Project Management
- Leadership and Management Skills
- Problem-Solving and Analytical Thinking
- Communication Skills
- Team Collaboration
- Time Management
- Computer Literacy
- Attention to Detail

LANGUAGES

- English (Proficient)
- Sinhala (Fluent)

EXTRA-CURRICULAR ACTIVITIES

The Open University of Sri Lanka - Nawala

- Student Representative (Faculty of Engineering) - 2023
- Coordinator of FETSAC (Faculty of Engineering Technology Student Academic Conference) - 2023

St. John's College - Nugegoda

- Member School Prefect Guild
- Member of the School Athletics Team

NON - RELATED REFEREES

- Kaveendra Sampath
Research Engineer
ACCIMT - Katubedda
Contact No : +94 772985395
- K. M. P. Koggala
Lecturer
The Open University of Sri Lanka
Contact No : +94 761646127

EDUCATIONAL QUALIFICATIONS

- Bachelor of Science Honours in Engineering (Special - Electronics & Communication Engineering) degree at The Open University of Sri Lanka
- G.C.E Advanced Level Examinations 2018
(St. John's College, Nugegoda / Physical Science)
- G.C.E Ordinary Level Examinations 2014
(St. John's College, Nugegoda)

CERTIFICATIONS

- Certificate Course in English Language at Institute of Human Resource Advancement - University of Colombo

PROJECTS

- Identification And Investigation Of Aflatoxin Presence And Varieties In Corn Flour
Technology and Features - STM32(Black pill)-High Frequency Synchronous Clk Pulse Generate , Sony ILX555K RGB CCD, GUVVA-S12SD UV Sensor , 16Bit I2C ADC , PK268E2 Stepper Motor , A4988 DRIVER , Scanning Mechanism (CNC), STMCUBE IDE/MX, ST LINK V2 USB, Firm Ware Configuration and Debug
- Railway Elephant Detection System - ACCIMT
Technology and Features - AI, Machine Learning ,YOLO, Thermal Camera , Nema23, DMD556 Motor Controller , Rain Sensor YL-83 FC-37, Up-Lift Mechanism (CNC)
- IOT Water Management System to University SNT Building
Technology and Features - ESP32, Arduino IDE, YF-S201 Water Flow Sensor, Solenoid, RTC , C++
- IoT Smart Gas Safety System
Technology and Features - Arduino ,C++ (Arduino IDE), MQ5 ,I2C Display, Solenoid, SIM 900A
- Portable Weather Station - ACCIMT
Technology and Features - ATmega328P, LoRa RA-02 SX1278 Transceiver, Solar, EC Humidity Temperature Soil Sensor RS485, Weather Station Anemometer, Liquid Level Measurement PVDF Ultrasonic Sea Water Level Sensor
- Obstacle Avoiding Robot - ACCIMT
Technology and Features - Arduino, HC-SR04 Ultrasonic Sensor, L298N Motor Driver, DC Motor, Servo Motor- SG90, C++, Arduino IDE
- Automatic Washing Machine
Technology and Features - Arduino

RESEARCH

- Limited Area Self Driving Vehicle (Ongoing)
Research Area - AI, Obstacle Detection, Machine learning, GPS Systems, Synchronous Fast Data Handling
- Hybrid Drone for Goods Delivery (Ongoing)
Research Area - Battery Pack Charging System Through a Small 2 Stroke Engine and Power handling, Motor RPM Handling Drone Weight and Gyroscope Readings, GPS System, RF Antenna